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Project Executable Files

Date	1 October 2026
Team ID	SWTID-2026-6912
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	3 Marks

Project Executable Files

This section contains the executable and deployable files required for FitBuddy. The project is a FastAPI web application integrated with Google Gemini models, SQLite/SQLAlchemy, and Jinja2 templates. The evaluator should be able to run the project using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes - SQLite / SQLAlchemy database
5	Environment configuration file (.env.example or similar)	Yes - Gemini API key configuration
6	Deployed application URL (if hosted)	NA - Local deployment
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA - Web application
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes - Local feature testing
10	Demo video or walkthrough (if required)	NA / To be added if required

Step 2: File / Folder Structure

Provide an overview of the FitBuddy project's file and folder structure below:

```
/FitBuddy-project

/app - FastAPI application source files
  main.py - application entry point
  routes.py - application routes and request processing
  gemini_generator.py - Gemini workout plan generation
  gemini_flash_generator.py - nutrition/recovery tip generation
  updated_plan.py - feedback-based plan updating
  database.py - SQLite/SQLAlchemy storage

/templates - Jinja2 HTML templates
  index.html - user input form
  result.html - workout plan, nutrition tip and feedback
  all_users.html - admin/user overview
/static/images - UI/image assets
/tests - Test files and test results (if included)
/docs - Project documentation
requirements.txt - Python dependencies
README.md - Setup and run instructions
.env.example - Gemini API key/environment configuration
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Local deployment: http://127.0.0.1:8000
Login Credentials (Demo)	No separate login required for the current application.
Platform / Hosting Provider	Local system using FastAPI + Uvicorn
Repository Link	To be provided by the team / GitHub repository
Demo Video Link	To be provided if required

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run the FitBuddy project locally:

Pre-requisites:
Python 3.x, pip, FastAPI, Uvicorn, Jinja2, SQLAlchemy, python-multipart and Google Generative AI SDK. A valid Gemini API key is required.

Step 1: Clone/download the FitBuddy project source code.
Step 2: Create and activate a virtual environment: python -m venv fitbuddy-env
  Windows: fitbuddy-env\Scripts\activate
Step 3: Install dependencies: pip install -r requirements.txt
Step 4: Configure the Gemini API key in the environment / .env file: GOOGLE_API_KEY=your_gemini_api_key_here
Step 5: Start the FastAPI application: uvicorn app.main:app --reload
Step 6: Open the application: http://127.0.0.1:8000
Step 7: API documentation: http://127.0.0.1:8000/docs

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Gemini API requires a valid API key and network access.	Configure GOOGLE_API_KEY before running; verify API access.
2	AI-generated plans depend on the quality of user input and model response.	Provide clear fitness goal and intensity; review generated plan.
3	Current setup is intended for local development/deployment.	Run with Uvicorn locally; cloud hosting can be added later.